# Initial raw data clean-up

## Overview

This notebook describes my initial exploration and cleaning of the raw data from the fall 2026 CURIOSS OSPO survey.

## Import packages, utilities, and config variables

```
library(readxl)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(docstring)
```

```
Attaching package: 'docstring'
```

```
The following object is masked from 'package:utils':

    ?
```

```
# Read config.yml
cfg <- config::get()
```

## Define functions

```
remove_uninformative_rows <- function(df) {
  #' Remove Uninformative Rows
  #'
  #' Removes rows that contain either an NA object, an empty string, or a
  #' whitespace string ("  ") in every column.
  #'
  #' @param df Data frame to process.
  #' @return A copy of the data frame with uninformative rows removed.
  df %>%
    filter(if_any(everything(), function(x) {
      val <- trimws(x) # trim whitespace
      !is.na(val) & val != ""
    }))
}

write_df_to_file <- function(df, filename) {
  #' Write data frame to a tsv file
  #'
  #' @description
  #' Exports a given data frame as a plain, tab-delimited text file saved within
  #' the directory specified by the `data_path` variable in config.yml.
  #' Include .tsv extension when specifying filename.
  #'
  #' @param df A data frame to be written to file.
  #' @param filename A character string specifying the destination file name.
  #'
  #' @return `NULL` (called for its side effect of writing a file).

  write.table(
    df,
    file.path(cfg$data_path, filename),
```

```
    quote = FALSE,
    row.names = FALSE,
    sep = "\t"
  )
}

select_and_rename <- function(df, name_map) {
  #' Select and Rename Columns Using a Named Vector
  #'
  #' @description
  #' Extracts a subset of columns from a data frame based on the values in a
  #' named vector and renames them to match the corresponding vector names.
  #'
  #' @param df A data frame or tibble containing the source columns.
  #' @param name_map A named character vector where values represent existing
  #' column names and keys/names represent the nicknames.
  #'
  #' @return A subsetted and renamed data frame (or tibble).
  #'
  #' @examples
  #' df <- data.frame(
  #'   "In which region is your organization located?" = c("NA", "EU"),
  #'   "Which best describes your role?" = c("Dev", "Mgr"),
  #'   check.names = FALSE
  #' )
  #'
  #' mapping <- c(
  #'   "region_mc" = "In which region is your organization located?",
  #'   "role_mc"   = "Which best describes your role?"
  #' )
  #'
  #' select_and_rename(df, mapping)

  result <- df %>%
    select(any_of(unname(name_map))) %>%
    rename(any_of(name_map))

  return(result)
}
```

## Read data from excel

After receiving the data from Clare, a password-protected .xlsx file, I copied the data into a new excel file without password protection so R's readxl package could read it. That file is called raw_data.xlsx.

```
data <- readxl::read_excel("data/raw_data.xlsx", col_types = "text", na = c(""))
# Using a workaround to preserve distinction between
# "" and "N/A", because by default read_excel converts
# both of these to NA objects.
data[is.na(data)] <- ""
data[data == "N/A"] <- NA
```

## Initial exploration and clean-up

```
dim(data)
```

```
[1]  36 198
```

```
names(data)[1:15]
```

```
 [1] "Response ID"
 [2] "Date submitted"
 [3] "Last page"
 [4] "Start language"
 [5] "Seed"
 [6] "Quota exit"
 [7] "Your organization / institution name"
 [8] "Your email"
 [9] "In which region is your organization located?"
[10] "In which region is your organization located? [Other]"
[11] "Which best describes the open source function at your organization?"
[12] "Which best describes the open source function at your organization? [Other]"
[13] "Where is your OSPO located within your university? [University library]"
[14] "Where is your OSPO located within your university? [Central administration (e.g., centr
[15] "Where is your OSPO located within your university? [Technology transfer, innovation, or
```

Remove uninformative columns

```
cols_to_remove <- c(
    "Response ID",
    "Date submitted",
    "Last page",
    "Start language",
    "Seed",
    "Quota exit")

data <- data %>% select(-any_of(cols_to_remove))
dim(data)
```

```
[1]  36 192
```

Remove uninformative rows

```
data <- remove_uninformative_rows(data)
dim(data)
```

```
[1]  26 192
```

Remove the very first response, which I suspect was a test by one of the survey organizers. The organization is listed as "dsad" and the email is "dsas", and the locations of those keys on a qwerty keyboard support my hypothesis that that was just some nonsense for a test run. The only question that that person answered was the "choose your top 3" drag-and-drop, so I assume an organizer was just testing that that question worked.

```
data <- data %>% slice(-1)
```

## Simple multiple-choice

I'll start with the simple multiple choice questions. I'll dump all of these in a single file called mc_simple.tsv.

```
# Rather than indexing these questions based on column number, I'll grab them
# by the exact column header text.
mc_simple_names_map <- c(
  "scope" = "Who does your OSPO primarily serve?",
  "age" = "How long has your OSPO been operating?",
  "staff_time" = "Roughly how much total staff time is formally spent on OSPO or OSPO-like a
```

```
  "volunteers" = "Beyond that, roughly how many people regularly spend time on OSPO activitie
  "expectations" = "Over the next two years, what do you expect to happen to your OSPO or OSI
  "funding_duration" = "What is the time horizon of your current funding or allocation?",
  "followup" = "If required, would you be willing to take part in an optional 45 to 60 minute
)

mc_simple <- select_and_rename(data, mc_simple_names_map)
names(mc_simple)
```

```
[1] "scope"            "age"              "staff_time"       "volunteers"
[5] "expectations"     "funding_duration" "followup"
```

```
head(mc_simple)
```

```
# A tibble: 6 x 7
  scope       age   staff_time volunteers expectations funding_duration followup
  <chr>       <chr> <chr>      <chr>      <chr>        <chr>            <chr>
1 <p>Beyond ~ 1 to~ 1 to 2 FTE 6-10 peop~ "Stay about~ "Ongoing or rec~ Yes
2 <p>Our own~ 1 to~ Less than~ 2 to 5 pe~ "Grow (more~ "No defined fun~ No
3 <p>The who~ 3 to~ 1 to 2 FTE None       "Shrink"     "Tied to a gran~ Yes
4 <p>Our own~ More~ Between 0~ 6-10 peop~ "Stay about~ "Multi-year (2 ~ Yes
5 <p>The who~ 1 to~ Less than~ 2 to 5 pe~ "Grow (more~ "No defined fun~ Yes
6 <p>The who~ More~ 2 to 5 FTE None       ""           ""               <NA>
```

```
write_df_to_file(mc_simple, 'mc_simple.tsv')
```

## Multiple choice + other

Now let's create a couple more files for the multiple choice questions that had an "other" write-in option. mc_complex.tsv will have just the multiple-choice part of these "complex" multiple choice questions. mc_other.tsv has the corresponding "other" responses where people wrote-in additional info.

```
mc_complex_names_map <- c(
  "region_mc" = "In which region is your organization located?",
  "function_mc" = "Which best describes the open source function at your organization?",
  "role_mc" = "Which best describes your role?",
  "metrics_mc" = "Does your OSPO currently track any metrics or indicators of its impact?"
)
```

```r
mc_other_names_map <- c(
  "region_other" = "In which region is your organization located? [Other]",
  "function_other" = "Which best describes the open source function at your organization? [O
  "role_other" = "Which best describes your role? [Other]",
  "metrics_other" = "Name one or two of the metrics you track."
)


mc_complex <- select_and_rename(data, mc_complex_names_map)
names(mc_complex)
```

```
[1] "region_mc"   "function_mc" "role_mc"     "metrics_mc"
```

```r
head(mc_complex)
```

```
# A tibble: 6 x 4
  region_mc     function_mc                                   role_mc  metrics_mc
  <chr>         <chr>                                         <chr>    <chr>
1 Europe        A formally established OSPO                   I lead ~ We track ~
2 Europe        An emerging OSPO (in a setup or pilot phase) I am a ~ Not yet, ~
3 North America A formally established OSPO                   I am a ~ We track ~
4 Europe        OSPO-like activities without a formal office I lead ~ We track ~
5 North America An emerging OSPO (in a setup or pilot phase) I lead ~ Not yet, ~
6 North America A formally established OSPO                   I lead ~ Yes, we h~
```

```r
mc_other <- select_and_rename(data, mc_other_names_map)
names(mc_other)
```

```
[1] "region_other"   "function_other" "role_other"     "metrics_other"
```

```r
head(mc_other)
```

```
# A tibble: 6 x 4
  region_other function_other role_other metrics_other
  <chr>        <chr>          <chr>      <chr>
1 ""           ""             ""         "The number of software programs devel~
2 ""           ""             ""         ""
3 ""           ""             ""         "Number of projects that have engaged ~
4 ""           ""             ""         "Number of published OSS repos, number~
5 ""           ""             ""         ""
6 ""           ""             ""         "Organizations and Repositories in the~
```

```
write_df_to_file(mc_complex, 'mc_complex.tsv')
write_df_to_file(mc_other, 'mc_other.tsv')
```

## Qualitative responses

Let's put the qualitative responses (not associated with a multiple-choice (MC) or select-all-that-apply (SATA) question) in their own file. I like to split obvious personally identifiable information (PII) into its own file, just to mitigate the potential bias introduced by looking at it while I'm reading the responses.

TODO!